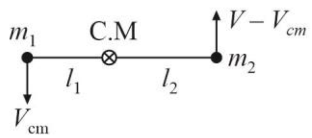
Solutions to JEE Advanced - 11 |JEE 2022 | Paper - 2

PHYSICS

1.(BD)

$$V_{cm} = \frac{m_2 V}{m_1 + m_2}$$



$$l_1 = \frac{m_2 l}{m_1 + m_2}$$

Both particles will rotate with ω about C.M.

$$\omega = \frac{V_{cm}}{l_1} = \frac{V}{l}$$

$$T = m_1\omega^2 l_1 = \frac{m_1 V^2}{l^2}\frac{m_2 l}{m_1+m_2} = \frac{m_1 m_2}{(m_1+m_2)}\frac{v^2}{l}$$

After time $\frac{\pi}{2\omega}$ strong will rotate by $\frac{\pi}{2}$

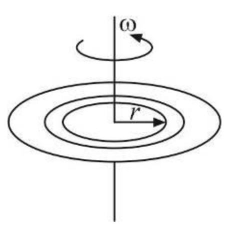
Displacement in $\frac{\pi l}{V}$ is

$$\sqrt{(2l_1)^2 + \left(\frac{\pi l}{V} \times \frac{m_2 V}{m_1+m_2}\right)^2} = \sqrt{\left(\frac{2m_2 l}{m_1+m_2}\right)^2 + \left(\frac{\pi l\, m_2}{m_1+m_2}\right)^2} = \frac{m_2 l}{m_1+m_2}\sqrt{4+\pi^2}$$

2.(AB) $F \times \frac{3R}{4} = \tau$

$$\int dp = \int \eta 2\pi r dr \times \frac{\omega r^2 \omega}{T}$$

$$P = \frac{\pi\omega^2 \eta R^4}{2T}$$



3.(ABD)

$$\rho(z) = \frac{\rho_0}{2}\left(1 + \frac{P(z)}{P_0}\right)$$

$$P(z) = \int \rho(z)\, gdz + P_{atm} = \int_0^x \frac{\rho_0}{2}\left(1+\frac{P(z)}{P_0}\right) gdz + P_{atm}$$

$$\frac{dP(z)}{dz} = \frac{\rho_0}{2}\left(1+\frac{P(z)}{P_0}\right) g$$

$$\log\left(\frac{P_0 + P(z))}{P_0 + P_{atm}}\right) = \frac{\rho_0 gz}{2P_0}$$

$$\Rightarrow \qquad P(z) = \left[(P_0 + P_{atm})\, e^{\rho_0 gz/2P_0 - P_0}\right]$$

$$\rho(z) = \rho_0 (P_0 + P_{atm})\, e^{\rho_0 gz/2P_0}$$

$$\rho(z) = \frac{\rho_0 (P_0 + P_{atm})}{2P_0}\, e^{\rho_0 gz/2P_0}$$

Mass of the liquid $M = \int_0^h \rho(z)\, Adz = \frac{A}{g}(P_0 + P_{atm})\,(e^{\rho_0 gh/2P_0} - 1)$

4.(ABCD)

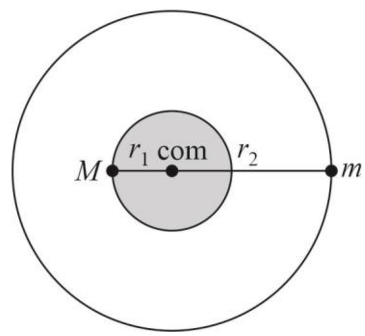
Given that M is the mass of the nucleus (proton) and m the mass of electron. Proton and electron both revolve about their centre of mass (COM) with same angular velocity (ω) but different linear speeds. Let r_1 and r_2 be the distance of COM from proton and electron.

Let r be the distance between the proton and the electron. Then

$$Mr_1 = mr_2$$

$$r_1 + r_2 = r$$

$$\therefore \quad r_1 = \frac{mr}{M+m} \text{ and } r_2 = \frac{Mr}{M+m}$$



Centripetal force to electron is provided by the electrostatic force.

So, $mr_2\omega^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$

or $\quad m\left(\frac{Mr}{M+m}\right)\omega^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$

or $\quad \left(\frac{Mm}{M+m}\right)r^3\omega^2 = \frac{e^2}{4\pi\varepsilon_0}$

Substituting $\frac{Mm}{M+m} = \mu$ (reduced mass of proton and electron)

$$\mu r^3\omega^2 = \frac{e^2}{4\pi\varepsilon_0} \qquad \ldots(1)$$

Moment of inertia of the atom about COM is :

$$I = Mr_1^2 + mr_2^2$$

or $\quad I = \left(\frac{Mm}{M+m}\right)r^2 = \mu r^2 \qquad \ldots(2)$

According to Bohr's theory $I\omega = n\hbar$ where $\left(\hbar = \frac{h}{2\pi}\right)$

or $\quad \mu r^2\omega = n\hbar \qquad \ldots(3)$

Solving equation (1) and (3) for r we get

$$r = \frac{4\pi\varepsilon_0 n^2\hbar^2}{\mu e^2} \qquad \ldots(4)$$

Electrical potential energy of the system is $U = -\frac{e^2}{4\pi\varepsilon_0 r}$

and kinetic energy is $K = \frac{1}{2} I\omega^2 = \frac{1}{2}\mu r^2 \omega^2$

But from equation (1)

$$\omega^2 = \frac{e^2}{4\pi\varepsilon_0 \mu r^3} \qquad \therefore K = \frac{e^2}{8\pi\varepsilon_0 r}$$

$\therefore$ Total energy of the atom is $E = K + U = -\frac{e^2}{8\pi\varepsilon_0 r}$

Substituting value of r from equation (4) we get

$$E = -\frac{\mu e^4}{32\pi^2 \varepsilon_0^2 n^2 \hbar^2}$$

For ground state $n = 1$

$$E_1 = -\frac{\mu e^4}{32\pi^2 \varepsilon_0^2 \hbar^2}$$

or binding energy $E_b = |E_1| = \frac{\mu e^4}{32\pi^2 \varepsilon_0^2 \hbar^2}$

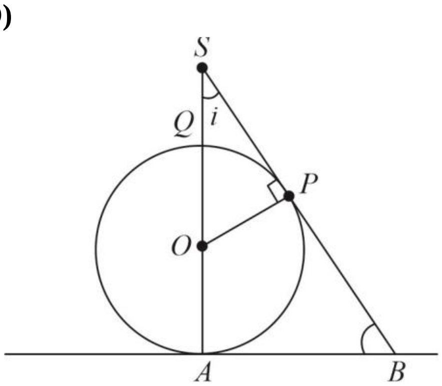
If motion of nucleus is not taken into account, in that case $E_b' = \frac{me^4}{32\pi^2 \varepsilon_0^2 \hbar^2}$

$$m > \mu \qquad \therefore \qquad E_b' > E_b$$

$\therefore$ Percentage increase $= \frac{E_b' - E_b}{E_b} \times 100 = \left(\frac{m-\mu}{\mu}\right) \times 100 = \left(\frac{m - \frac{mM}{M+m}}{\frac{mM}{M+m}}\right) \times 100$

$$= \left(\frac{m}{M}\right) \times 100 = 0.00055 \times 100 = 0.055\%$$

5.(ABD)



Given, $SQ = OQ = OA = R$

If $i =$ angle which a tangential light ray makes with line SA, then

$$\sin i = \frac{OP}{OS} = \frac{R}{2R} = \frac{1}{2} \quad \Rightarrow \quad \cos i = \sqrt{1^2 - \left(\frac{1}{2}\right)^2} = \frac{\sqrt{3}}{2}$$

$\therefore \qquad \cot i = \sqrt{3} \qquad \ldots\text{(i)}$

Now, in ΔSAB,

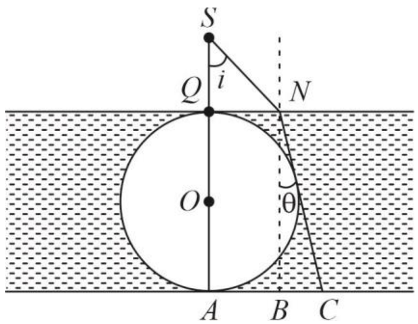
$\tan(90-i)=\tan\ \angle SBA$

$$=\frac{SA}{AB}=\frac{3R}{r}\quad \text{(where, } r=AB= \text{ radius of shadow)}$$

$\Rightarrow \quad \cot i=\sqrt{3}=\dfrac{3R}{r} \qquad [\because \text{ from Eq. (i)}]$

$\Rightarrow \quad r=\sqrt{3}\ R$

$\therefore$ Shadow has an area of $\pi r^2=3\pi R^2$ sq units.



Let a light ray is refracted at N and is tangential to sphere.

Also, $\dfrac{\sin i}{\sin r}=\mu$

$\Rightarrow \quad \dfrac{\sin i}{\sin\theta}=\sqrt{3} \quad \Rightarrow \quad \sin\theta=\dfrac{\sin i}{\sqrt{3}}$

Then, radius of shadow

$$AC=AB+BC$$

$$r'=AC=2R\tan\theta+R\tan i$$

Also, $r'=2R\sec\theta-R\tan i$

So, $2R\sec\theta-R\tan i=2R\tan\theta+R\tan i$ or $\sec\theta-\tan\theta=\tan i$

$$\Rightarrow \quad \frac{1-\sin\theta}{\cos\theta}=\tan i \quad \Rightarrow \quad \frac{\left(\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\right)^2}{\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}}=\tan i \quad \Rightarrow \quad \frac{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}+\sin\frac{\theta}{2}}=\tan i$$

$$\Rightarrow \quad \frac{1-\tan\frac{\theta}{2}}{1+\tan\frac{\theta}{2}}=\tan i \quad \Rightarrow \quad \tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)=\tan i \quad \Rightarrow \quad \frac{\pi}{4}-\frac{\theta}{2}=i \quad \Rightarrow \quad \frac{\pi}{2}-\theta=2i$$

Now, using $\sin\theta=\dfrac{\sin i}{\sqrt{3}}$, we get

$$\Rightarrow \quad \sin\left(\frac{\pi}{2}-2i\right)=\frac{\sin i}{\sqrt{3}} \quad \Rightarrow \quad \cos 2i=\frac{\sin i}{\sqrt{3}}$$

$$1-2\sin^2 i=\frac{\sin i}{\sqrt{3}}$$

$$2\sqrt{3}\sin^2 i+\sin i-\sqrt{3}=0 \quad \Rightarrow \quad \sin i=\frac{1}{\sqrt{3}} \text{ or } -\frac{\sqrt{3}}{2}$$

In given case angle i is acute

So, $\sin i = \frac{1}{\sqrt{3}}$

Hence, $\sin\theta = \frac{\sin i}{\sqrt{3}} = \frac{1}{3}$

So, we have $r' = 2R\tan\theta + R\tan i = 2R \times \frac{1}{\sqrt{8}} + R \times \frac{1}{\sqrt{2}} = R\sqrt{2}$

6.(AD) Let *A* and *B*, the number of α and β particles are emitted when ${}^{238}_{92}U$ decay to ${}^{206}_{82}Pb$.

We know that

(i) The emission of a $\alpha-$particle $({}^{4}_{2}He)$ decreases the charge number by two and mass number by four

Thus emission of *a*, α-particles reduce the charge number by 2*a* and mass number by 4*a*.

(ii) The emission of β-particle increase the charge number by one and leaving the mass number unchanged.

Thus emission of *b*, β-particles increases the charge number by $b \times 1 = b$.

Thus, ${}^{238}_{92}U \to {}^{206}_{82}Pb + a({}^{4}_{2}He) + b({}^{0}_{-1}\beta)$

Applying the law of conservation of charge number and mass number before and after the decay, we have

$92 = 82 + 2a - b$ and $238 = 206 + 4a$

Solving, $a = 8$; $b = 6$

7.(12.40)

The maximum kinetic energy of the electrons immediately upon ejection is the difference between the energy of the incident photon and the threshold energy.

$$K = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

This kinetic energy of ejected electron is converted to electrostatic potential energy, $\Delta U = eEd$, as electrons come to rest while moving in the direction of electric field. Therefore, $K = Eed$.

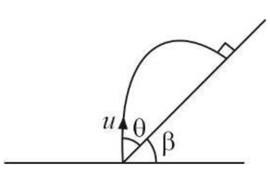
and $\lambda_0 = \left(\frac{1}{\lambda} - \frac{eEd}{hc}\right)^{-1} \Rightarrow \lambda_0 = 2\lambda \Rightarrow d = \frac{hc}{2\lambda eE} = 12.43\, m$

8.(3.5) For ball to retrace path, it must hit the inclined plane at 90°

$\therefore \quad v_x = 0 \to 2\tan\theta\tan\beta = 1$

$$R = u_x T - \frac{1}{2} g \sin\beta T^2$$

On solving $u = \sqrt{\frac{gR(1+3\sin^2\beta)}{2\sin\beta}}$



Put in the values, $u = 3.5\, m/s$

9.(65) Let the resistance of the coils be R_A *and* R_B

Let the heat capacity of 1 kg of water be C

Then, we are given that

$$\frac{V^2}{R_A}=\frac{C(60-20)}{10(60)} \qquad and \qquad \frac{V^2}{R_B}=\frac{C(80-20)}{5(60)}$$

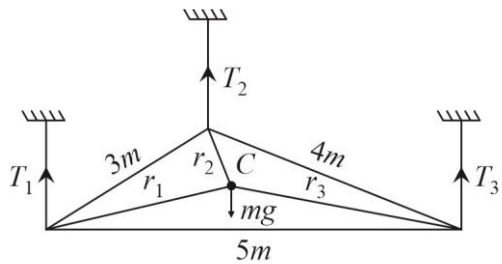
So, $\quad R_A=\dfrac{15V^2}{C} \qquad and \qquad R_B=\dfrac{5V^2}{C}$

In series, let the final temperature of water be T

Then, $\quad \dfrac{V^2}{R_A+R_B}=\dfrac{C(T-20)}{15(60)} \quad \Rightarrow \quad \dfrac{V^2}{\dfrac{15V^2}{C}+\dfrac{5V^2}{C}}=\dfrac{C(T-20)}{900} \quad \Rightarrow \quad T=65°C$

10.(26.67)

$T_1+T_2+T_3=mg \qquad \ldots(1)$



Balancing torque about C

$\vec{r}_1\times\vec{T}_1+\vec{r}_2\times\vec{T}_2+\vec{r}_3\times\vec{T}_3 \qquad \ldots(2)$

and $\quad \vec{r}_1+\vec{r}_2+\vec{r}_3=0 \qquad \ldots(3)$

From (1), (2) & (3)

$$T_1=T_2=T_3=\frac{mg}{3}=\frac{40}{3}$$

11.(2.5)

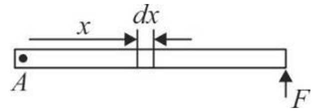
$$\varepsilon=-a^2\frac{dB}{dt}$$

$$\frac{dB}{dt}=\frac{vdB}{dx}+\frac{dB}{dt}$$

$$E=-\frac{1}{4}(-2\times10^{-3}+10^{-3})=\frac{1}{4}\times10^{-3}$$

$$i=\frac{0.25\times10^{-3}}{0.1\times10^{-3}}=2.5A$$

12.(3) When hinged at A



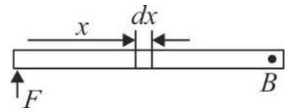
$$I_A=\int(dm)x^2=\int_0^{\ell}\lambda_0 x\,dx\,x^2=\frac{\lambda_0\ell^4}{4}$$

$$\tau_A=I_A\alpha_A \quad \Rightarrow \quad \alpha_A=\frac{F\ell}{\left(\dfrac{\lambda_0\ell^4}{4}\right)}=\frac{4F}{\lambda_0\ell^3}$$

When hinged at B

$$I_B = \int (dm)(\ell - x)^2$$

$$= \int_0^{\ell} \lambda_0 x\, dx(\ell^2 + x^2 - 2\ell x) = \lambda_0 \left[\ell^2 \frac{\ell^2}{2} + \frac{\ell^4}{4} - 2\ell \frac{\ell^3}{3} \right] = \frac{\lambda_0 \ell^4}{12}$$



13.(19)

By cons of momentum $2mu - mu = 3mv$

$$V = \frac{1}{3}\sqrt{\frac{GM}{2R}}$$

Total energy $= \frac{P^2}{6m} - \frac{GM\,(3m)}{R}$

Where P is linear momentum when it hits the earth

$$\frac{P^2}{6m} - \frac{3GMm}{R} = \frac{-17GMm}{12r}$$

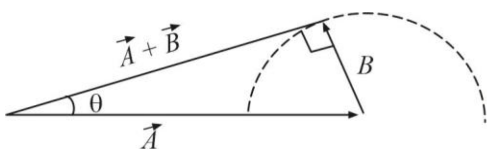
$$p = \frac{19}{2} m \sqrt{\frac{GM}{R}}$$

By conservation of angular momentum

$$R_P \cos\theta = 3mv\ (2R)$$

$$\cos\theta = \frac{2}{\sqrt{2}} \times \frac{2}{19} = \frac{2\sqrt{2}}{19}; \qquad \sec\theta = \frac{19}{2 \times 1.414} = 6.72$$

14.(12)



$$\sin\theta = \frac{10}{20} = \frac{1}{2} \quad \Rightarrow \quad \theta = 30^\circ$$

Angle between $\vec{A}$ & $\vec{B}$ is $120^\circ = 10^\circ \times 12$

15.(B) Standard current carrying distributions can be used to calculate dependence.

16.(A) Word done is positive in clockwise process and negative in anti-clockwise process.

17.(C)

18.(B) **(P)** $\vec{v} = \alpha\hat{i} + \beta j$

Speed is constant. Particle moves in straight line passing through origin, $\omega = 0$ always.

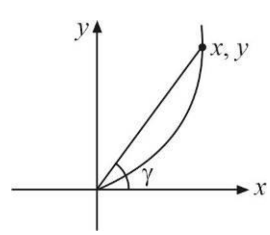
(Q) Uniform circular motion both v and v and ω are constant.

(S) $x = \alpha t,\ y = \frac{\beta t^2}{2}$

$y = \frac{\beta}{2\alpha^2} x^2 \to$ parabolic path

$\vec{v} = \alpha\hat{i} + \beta t\, j \to$ speed is increasing

$\omega = \frac{d\alpha}{dt}$



$$\tan\gamma = \frac{y}{x} = \frac{\beta t}{2\alpha}$$

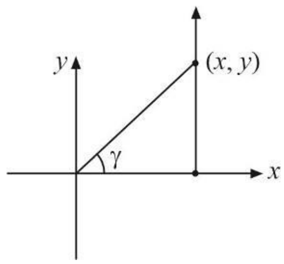
$$\sec^2\gamma\frac{d\alpha}{dt} = \frac{\beta}{2\alpha}$$

$$\omega = \frac{\beta}{2\alpha}\cos^2\gamma = \frac{2\beta\alpha}{\sqrt{\beta^2t^2+4\alpha^2}}$$

ω decreases with time.

(R) $\vec{v} = \beta j$

$$\tan\gamma = \frac{y}{x} = \frac{\beta t}{\alpha}$$



$$\sec^2\gamma\frac{dx}{dt} = \frac{\beta}{\alpha}$$

$$\omega = \frac{\beta}{\alpha}\cos^2\gamma = \frac{\beta}{\sqrt{\alpha^2+\beta^2t^2}}$$

ω decreases with time

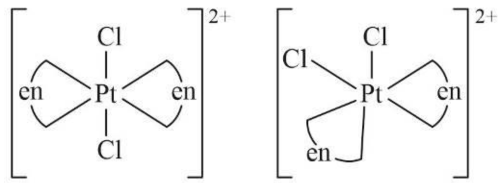
CHEMISTRY

1.(ACD)

Let oxidation number of Pt = x

$1 \times x + 2 \times 0 + 2 \times (-1) = +2$

$\Rightarrow \qquad x = +4$



Trans form (Optically inactive) cis form (Optically active)

2.(BC) $Ag^{+} + HCl \longrightarrow AgCl\downarrow + H^{+}$

The precipitate dissolve in concentrated HCl, $AgCl\downarrow + Cl^{-} \longrightarrow AgCl_2^{-}$

When KCN is added in small amount the Cu^{2+} ions first form a yellow precipitate of $Cu(CN)_2$ which on standing decomposes to white precipitate of CuCN.

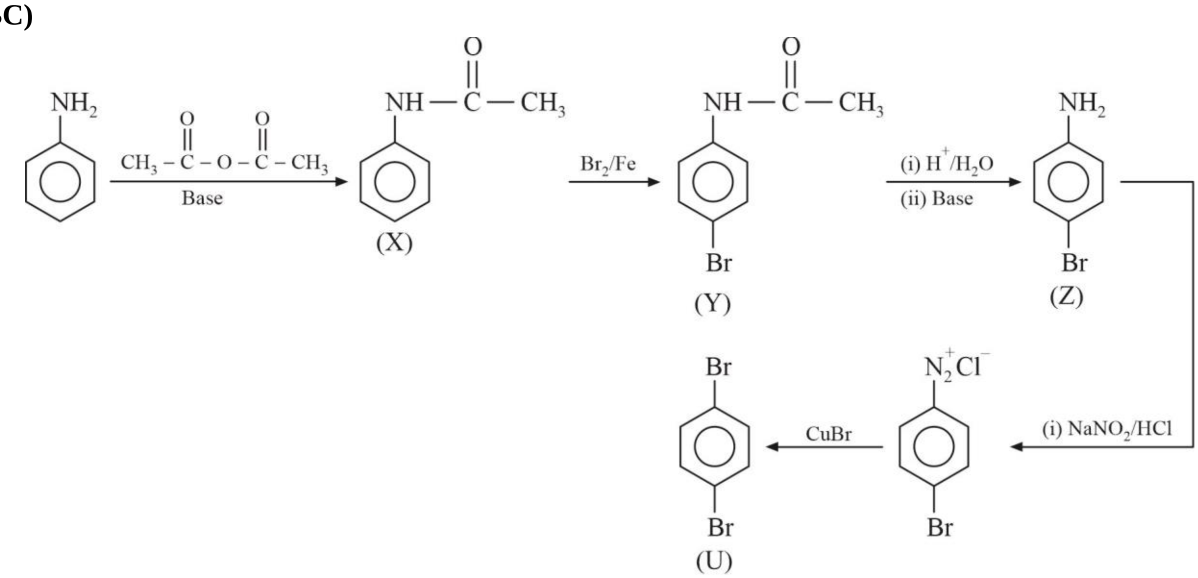
$$Cu^{2+} + 2CN^{-} \longrightarrow \underset{\text{(yellow ppt.)}}{Cu(CN)_2\downarrow}$$

$$\underset{\text{yellow}}{2Cu(CN)_2} \longrightarrow \underset{\text{white}}{2CuCN\downarrow} + (CN)_2$$

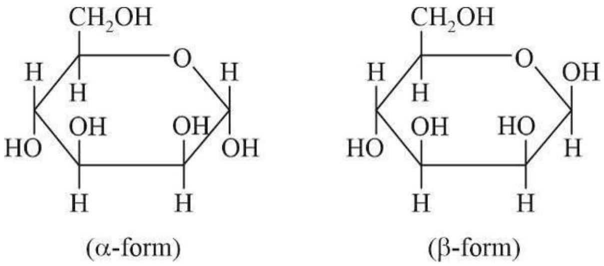
$$Hg^{2+} + \underset{\underset{(CH_3COO)_2Co}{\uparrow}}{Co^{2+}} + \underset{\underset{NH_4SCN}{\uparrow}}{4SCN^{-}} \longrightarrow \underset{\text{Blue precipitate}}{Co[Hg(SCN)_4]}$$

$$\underset{\text{Black precipitate}}{BiI_3\downarrow} + H_2O \longrightarrow \underset{\text{Orange}}{BiOI\downarrow} + 2H^{+} + 2I^{-}$$

3.(ABC)



4.(BD) The structure of D-Mannose is



5.(CD) For first order kinetics, rate = KC

$$C = C_0 e^{-kt}$$

$$k = \frac{2.303}{t} \log \frac{C_0}{C}$$

At $\quad t = t_{3/4},\ C = \frac{C_0}{4}$

$$t_{3/4} = \frac{2.303}{k} \times \log \frac{C_0}{C_{0/4}}$$

$$t_{3/4} = \frac{2.303}{k} \times \log 4$$

$$t_{3/4} = \frac{1.386}{k}$$

$$-\log C + \log C_0 = \frac{k}{2.303} \cdot t$$

$$\log C = -\frac{k}{2.303} \cdot t + \log C_0$$

6.(ABCD)

$$T_i = \frac{P_i V_i}{nR}$$

$$T_i = \frac{1 \times 1}{1 \times R} = \frac{1}{R}$$

$$T_f = \frac{P_f V_f}{nR}$$

$$T_f = \frac{1 \times 2}{1 \times R} = \frac{2}{R}$$

$$T_f = 2T_i$$

$$\Delta U = nC_V\ \Delta T = 1 \times \frac{3}{2} \times R \times (T_f - T_i)$$

$$\Delta U = \frac{3}{2} \times R \times T_i = \frac{3}{2} \times P_i V_i$$

$$\Delta U = \frac{3}{2} \times 1 \times 1 = \frac{3}{2} \text{ litre bar} = \frac{3}{2} \times 100 \text{ Joule}$$

$$\Delta U = 150 \text{ Joule}$$

H = U + PV

$$\Delta H = \Delta U + \Delta(PV) = \frac{3}{2}P_iV_i + (P_fV_f - P_iV_i)$$

$$\Delta H = \frac{3}{2}P_iV_i + P_iV_i$$

$$\Delta H = \left(\frac{3}{2}+1\right) \text{ litre bar}$$

$$\Delta H = \frac{5}{2}\times 100 \text{ J}$$

$$\Delta H = 250 \text{ J}$$

Entropy change is positive

Work done by gas is zero

7.(4) SO_2, SO_3, H_3PO_4 and $HClO_4$ has $p\pi - d\pi$ bonding.

8.(130) Moles of $Na[Ag(CN)_2] = \frac{500}{1000}\times 0.5 = 0.25 = \frac{1}{4}$

$$\underset{\frac{1}{4}}{Zn + 2Na[Ag(CN)_2]} \longrightarrow Na_2[Zn(CN)_4] + 2Ag$$

Moles of Zn required $= \frac{1}{4}\times\frac{1}{2} = \frac{1}{8}$

Mass of Zn required = x $= \frac{1}{8}\times 65 = \frac{65}{8}$ gram

Hence $16x = \frac{65}{8}\times 16 = 130$

9.(110) $CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + Cl_2$

$$Cl_2 + 2KI \longrightarrow 2KCl + I_2$$

$$I_2 + 2Na_2S_2O_3 \longrightarrow Na_2S_4O_6 + 2NaI$$

m.eq of $Na_2S_2O_3 = 0.125\times 20 = 2.5$

m.eq of I_2 = M.eq of $Na_2S_2O_3 = 2.5$

m.eq of Cl_2 = m.eq of $I_2 = 2.5$

m.eq of Cl_2 in 100 ml solution $= 2.5\times\frac{100}{25} = 10$

Mass of chlorine $= \frac{10}{1000}\times\frac{71}{2}$ gram $= 0.355$ gram

% of chlorine = x $= \frac{0.355}{3.55}\times 100 = 10$

Hence 11x = 110

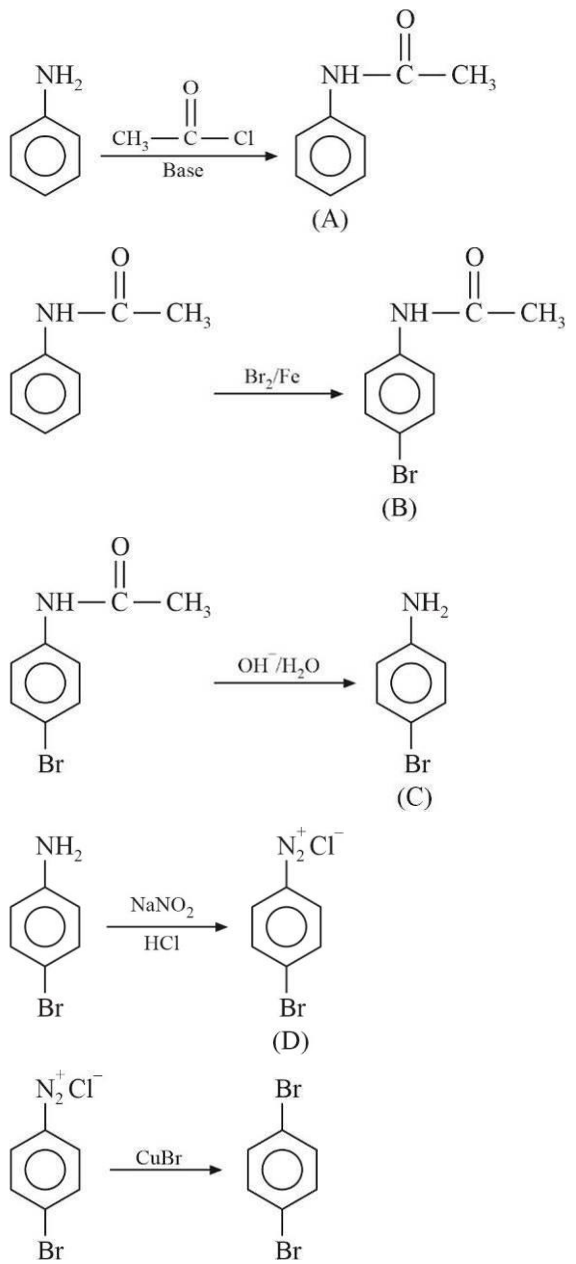
10.(256)

Number of stereo centres n = 4

Hence, number of stereo isomers $x = 2^n = 2^4 = 16$

Therefore, 16x = 16 × 16 = 256

11.(236)



Moles of aniline = 25

Moles of A = 25 × 0.4 = 10

Moles of B = 10 × 0.5 = 5

Moles of C = 5 × 0.4 = 2

Moles of D = 2 × 0.5 = 1

Moles of E = 1

Mass of E = 1 × 236 = 236 gram

12.(50) $NO_{(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons NO_{2(g)}$

$\Delta G^\circ = \Delta G_f^\circ(NO_2)_{(g)} - \Delta G_f^\circ(NO)_{(g)}$

$= 52 - 87$

$\Delta G^\circ = -35\ kJ$

$\Delta G^\circ = -2.303\ RT\ \log K_P$

$-35 \times 1000 = -2.303 \times 8.314 \times 298\ \log K_P$

$\Rightarrow \quad \log K_P = 6.13 \quad \Rightarrow \quad K_P = 1.34 \times 10^6$

	$NO_{(g)}$	$+\ \frac{1}{2}O_{2(g)}$	$\rightleftharpoons$	$NO_{2(g)}$
Initially	1	11/4		0
Equilibrium	p	9/4		1

$$K_p = 1.34 \times 10^6 = \frac{p_{NO_2}}{p_{NO} \times p_{O_2}^{1/2}} \quad \Rightarrow \quad 1.34 \times 10^6 = \frac{1}{p \times \left(\frac{9}{4}\right)^{1/2}}$$

$\Rightarrow \quad p = 5 \times 10^{-7} \quad \Rightarrow \quad p = 50 \times 10^{-8}$

Hence, $x = 50$

13.(1733)

$$A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B$$

	A	B
$t = 0$	$[A]_0$	0
$t = t$	$[A]_0 - x$	x
$t = t_{eq}$	$[A]_0 - x_{eq}$	x_{eq}

$[A]_{eq} = [A]_0 - x_{eq}$

$[B]_{eq} = x_{eq}$

$$(k_f + k_b) = \frac{1}{t}\ln\left(\frac{x_{eq}}{x_{eq} - x}\right)$$

$$(k_f + k_b) = \frac{6}{\ln 2}\ln\frac{[B]_{eq}}{[B]_{eq} - \frac{[B]_{eq}}{2}}$$

$k_f + k_b = 6 \qquad \ldots\ldots(i)$

$k_b - k_f = 2 \qquad \ldots\ldots(ii)$

On solving (i) and (ii)

$k_b = 4,\ k_f = 2$

$$K_{eq} = \frac{k_f}{k_b} = \frac{1}{2}$$

$$\Delta G^\circ = -RT \ln K_{eq}$$

$$\Delta G^\circ = -2.303\,RT \log K_{eq}$$

$$\Delta G^\circ = -2.303 \times 2500 \log \frac{1}{2}$$

$$\Delta G^\circ = 1733 \text{ Joule}$$

14.(30.88)

$$E^\circ_{25^\circ C} = 0.3525 \text{ V}$$

$$E^\circ_{20^\circ C} = 0.3523 \text{ V}$$

$$\frac{dE}{dT} = \frac{E^\circ_{25^\circ C} - E^\circ_{20^\circ C}}{(T_2 - T_1)} = \frac{0.3525 - 0.3533}{(25 - 20)}$$

$$\frac{dE}{dT} = -1.6 \times 10^{-4} \text{ volt/K}$$

$$\Delta S^\circ = nF\frac{dE}{dT} = 2 \times 96500 \times (-1.6 \times 10^{-4}) \text{ J/K}$$

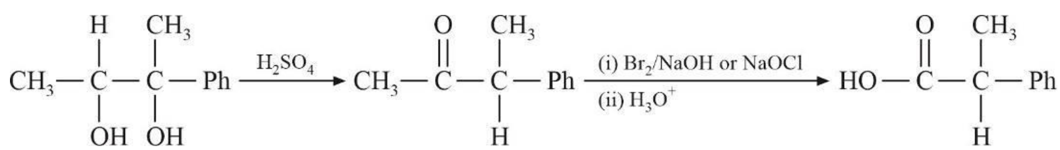
$$\Delta S^\circ = -30.88 \text{ J/K}$$

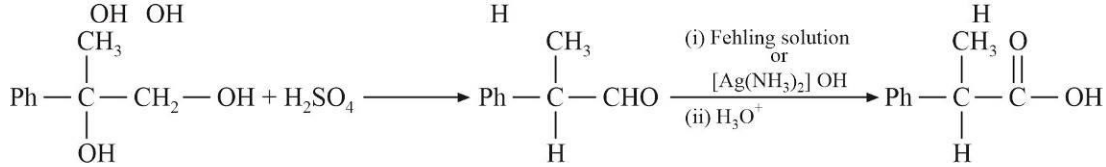
15.(B) $d^2sp^3 \rightarrow [Fe(en)_3]^{3+}$, $[Co(C_2O_4)_3]^{3-}$

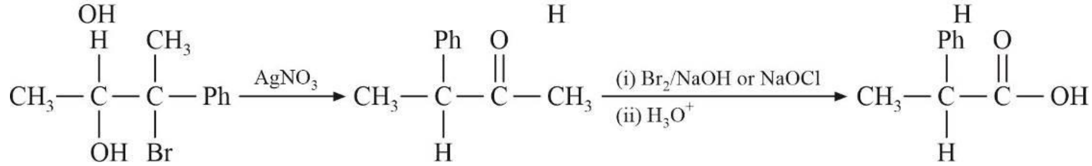
$sp^3 \rightarrow Na_2[Zn(CN)_4]$, $Ni(CO)_4$

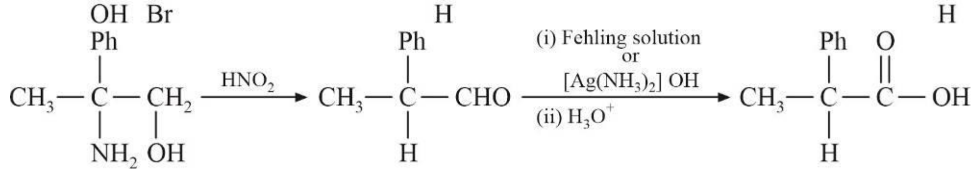
$sp^3d^2 \rightarrow [NiCl_6]^{2-}$

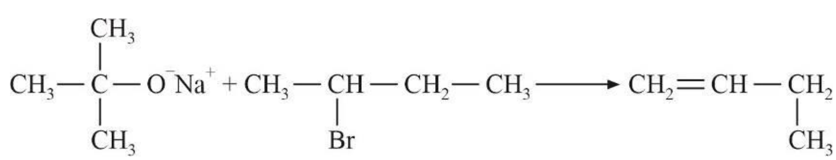
$dsp^2 \rightarrow K_2[PtCl_4]$

16.(C) **(P)** 

(Q) 

(R) 

(S) 

17.(A) **(P)** 

(Q) $CH_3CH_2CH_2CH_2OH \xrightarrow{H_2SO_4/\Delta} CH_3-CH=CH-CH_3$

(R) $CH_3-\underset{\underset{CH_3}{|}}{CH}-O-CH_2-CH_3 + \underset{(1\ mol)}{HI} \longrightarrow CH_3-\underset{\underset{CH_3}{|}}{CH}-OH + CH_3CH_2I$

(S) $CH_3-\underset{\underset{CH_3}{|}}{CH}-OH + HI \longrightarrow CH_3-\underset{\underset{CH_3}{|}}{CH}-I$

18.(C) **(P)**

	NH_4OH	+	HCl	$\longrightarrow$	NH_4Cl	+	H_2O
Initially	20		10		–		
Finally	10		0		10		

$pOH = pK_b$

$pH = 14 - pK_b$

Hence $[H^+]$ concentration does not change on dilution.

(Q)

	CH_3COOH	+	$NaOH$	$\longrightarrow$	CH_3COONa	+	H_2O
Initially	20		20		–		
Finally	0		0		20		

$$[OH^-] = \sqrt{\frac{C\,K_w}{K_a}}$$

$$[H^+] = \frac{K_w}{[OH^-]} = \sqrt{\frac{K_w K_a}{C}}$$

(R)

	CH_3COOH	+	$NaOH$	$\longrightarrow$	CH_3COONa	+	H_2O
Initially	20		40		–		
Finally	0		20		20		

Hence $[OH^-]$ will be controlled by NaOH

(S)

	NH_4OH	+	HCl	$\longrightarrow$	NH_4Cl	+	H_2O
Initially	20		40		–		
Finally	0		20		20		

Hence $[H^+]$ is controlled by HCl

MATHEMATICS

1.(AD) $f(n)=\sum_{r=1}^{n}\tan^{-1}\left(\frac{2r}{r^4+r^2+2}\right)=\sum_{r=1}^{n}\tan^{-1}\left(\frac{(r^2+r+1)-(r^2-r+1)}{1+(r^2+r+1)(r^2-r+1)}\right)$

$$=\sum_{r=1}^{n}\left[\tan^{-1}(r^2+r+1)-\tan^{-1}(r^2-r+1)\right]$$

$$f(n)=\tan^{-1}(n^2+n+1)-\frac{\pi}{4}$$

$$\lim_{n\to\infty} f(n)=\frac{\pi}{2}-\frac{\pi}{4}=\frac{\pi}{4}$$

$$\lim_{n\to\infty} f\left(\frac{1}{n}\right)-\lim_{n\to\infty} f(2n)$$

$$\lim_{n\to\infty}\left(\tan^{-1}\left(\frac{1}{n^2}+\frac{1}{n}+1\right)-\frac{\pi}{4}\right)-\lim_{n\to\infty}\left(\tan^{-1}(4n^2+2n+1)-\frac{\pi}{4}\right)$$

$$=\left(\frac{\pi}{4}-\frac{\pi}{4}\right)-\left(\frac{\pi}{2}-\frac{\pi}{4}\right)=-\frac{\pi}{4}$$

$\because \quad f(n)=\tan^{-1}(n^2+n+1)-\frac{\pi}{4}$

$$\lim_{n\to\infty}\left[\tan^{-1}(n^2+n+1)-\frac{\pi}{4}\right]$$

$$\lim_{n\to\infty}\left[\frac{\pi}{2}-\frac{\pi}{4}\right]=\lim_{n\to\infty}\left[\frac{\pi}{4}\right]=0$$

$$\tan\left(f(n)+\frac{\pi}{4}\right)=n^2+n+1$$

$$\cot\left(f(n)-\frac{\pi}{4}\right)=\cot\left(\tan^{-1}(n^2+n+1)-\frac{\pi}{2}\right)$$

$$=-\tan\tan^{-1}(n^2+n+1)=-(n^2+n+1)$$

2.(ACD)

$S_1: x^2+y^2+4y-1=0$ centre $C_1(0,-2)$; $r_1=\sqrt{5}$

$S_2: x^2+y^2+6x+y+8=0$ centre $C_2\left(-3,-\frac{1}{2}\right)$; $r_2=\frac{\sqrt{5}}{2}$

$S_3: x^2+y^2-4x-4y-37=0$ centre $C_3(2,2)$; $r_3=3\sqrt{5}$

$C_1C_2=r_1+r_2$ $\quad$ S_1 and S_2 touch externally

$C_2C_3=r_3-r_2$ $\quad$ S_2 and S_3 touch internally

$C_1C_3=r_3-r_1$ $\quad$ S_1 and S_3 touch internally

Point of contact of S_1 and S_2 is $P_1(-2,-1)$

Point of contact of S_2 and S_3 is $P_2(-4,-1)$

Point of contact of S_1 and S_3 is $P_3(-1,-4)$

Common tangent at $P_1 \Rightarrow 2x-y+3=0$

Common tangent at $P_2 \Rightarrow 2x+y+9=0$

Common tangent at $P_3 \Rightarrow x+2y+9=0$

Intersection of common tangents of circles (pairwise) is (–3, –3)

3.(AD) For given system of equations $\Delta=0$

$\Delta_1 = a + 7b - 13c$

$\Delta_2 = a + 7b - 13c$

$\Delta_3 = 0$

For atleast one solution ;

$a + 7b - 13c = 0 \qquad \ldots\ldots(i)$

Option A: $\Delta = 0$; $x + 2y + z = a$

$x + 2y + z = b/2$

$x + 2y + z = c/3$

For atleast one solution

$$a = \frac{b}{2} = \frac{c}{3}$$

But this relation is not satisfying equation (i)

Option B: $\Delta \neq 0$ consistent system

Option C: $\Delta \neq 0$ consistent system

Option D: $\Delta = 0$; $a = -\frac{b}{2} = 2c$

But not satisfying equation (i)

$\therefore$ System is inconsistent

4.(ABC)

Let $A(at_1{}^2, 2at_1)$; $B(at_2{}^2, 2at_2)$

$$\therefore \quad -t_1 - \frac{2}{t_1} = -t_2 - \frac{2}{t_2}$$

$t_1 t_2 = 2$

$$M = \left(\frac{a(t_1{}^2 + t_2{}^2)}{2}, a(t_1 + t_2)\right)$$

Let $C(h, k)$

$h = 2a + a(t_1^2 + t_1 t_2 + t_2^2) = 4a + a(t_1^2 + t_2^2)$

$k = -at_1 t_2 (t_1 + t_2) = -2a(t_1 + t_2)$

$$\text{Mid-point of } MC = (\alpha, \beta) = \left(\frac{\frac{a(t_1^2 + t_2^2)}{2} + h}{2}, \frac{a(t_1 + t_2) + k}{2}\right)$$

$$2\alpha = \frac{a(t_1^2 + t_2^2)}{2} + h,\ 2\beta = a(t_1 + t_2) + k = a(t_1 + t_2) - 2a(t_1 + t_2)$$

$$2\alpha = \frac{a(t_1^2 + t_2^2)}{2} + 4a + a(t_1^2 + t_2^2)$$

$$= \frac{3}{2}a(t_1^2 + t_2^2) + 4a \qquad\qquad 2\beta = -a(t_1 + t_2)$$

$$= \frac{3}{2}a[(t_1 + t_2)^2 - 2t_1 t_2] + 4a \qquad\qquad t_1 + t_2 = -\frac{2\beta}{a}$$

$$= \frac{3}{2}a\left[\frac{4\beta^2}{a^2} - 4\right] + 4a$$

$$2\alpha = \frac{6\beta^2}{a} - 6a + 4a$$

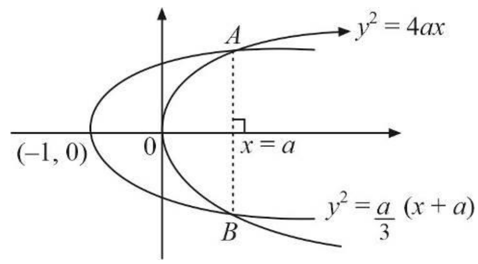
$$2\alpha = \frac{6\beta^2}{a} - 2a$$

$$\alpha = \frac{3\beta^2}{a} - a$$

Locus of (α, β) is

$$x = \frac{3y^2}{a} - a$$

$$\frac{a}{3}(x+a) = y^2$$



$$\frac{a}{3}(x+a) = 4ax \Rightarrow \quad x + a = 12$$

$$x = \frac{a}{11}$$

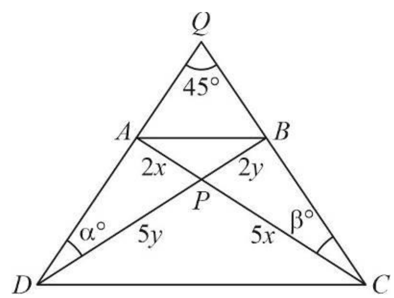
5.(ABC)

ΔABP and ΔCDP are similar,

let $AP = 2x$ and $BP = 2y$

then $CP = 5x$ and $DP = 5y$

Area of trapezium $ABCD = \frac{49}{2}xy$



$$\tan\alpha = \frac{2x}{5y}, \quad \tan\beta = \frac{2y}{5x}$$

$$\alpha + \beta = 45°$$

$$\Rightarrow \quad \frac{10(x^2+y^2)}{21xy} = 1$$

$$\Rightarrow \quad 10(x^2+y^2) = 21xy$$

Also, $\quad AB^2 = AP^2 + BP^2$

$$x^2 + y^2 = 4$$

$$\therefore \quad xy = \frac{40}{21}$$

Hence, Area of $ABCD$ $\quad = \frac{49}{2} \times \frac{40}{21} = \frac{140}{3}$

Area of triangle PCB $\quad = \frac{1}{2} \cdot 2y\, 5x = 5xy = \frac{200}{21}$

Difference of CP and PD is $= 5|x - y| = \frac{10}{\sqrt{21}}$

6.(ABC)

By LH rule

$$\lim_{t \to x} \frac{e^t f(x) - e^x f'(t)}{(f(x))^2} = 3$$

$$\frac{e^x f(x) - e^x f'(x)}{f^2(x)} = 3$$

$$d\left(\frac{e^x}{f(x)}\right) = 3$$

Integrate

$$\frac{e^x}{f(x)} = 3x + c$$

$\because \quad f(0) = 1$

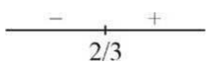
$$\frac{1}{1} = c \Rightarrow c = 1$$

$$\therefore \quad \frac{e^x}{f(x)} = 3x + 1 \qquad \therefore \quad f(x) = \frac{e^x}{3x+1}$$

$$f'(x) = \frac{(3x+1)e^x - 3e^x}{(3x+1)^2}$$

$$f'(x) = \frac{e^x(3x-2)}{(3x+1)^2}$$

$$f'(0) = -2$$



In, $x \in (1, 2)\, f(x)$ increasing function

$$f(x) > f(1)$$

$$\frac{e^x}{3x+1} > \frac{e}{4}$$

$$e^{x-1} > \frac{3x+1}{4}$$

7.(4) $$\int_0^{\pi/4} \frac{1}{\sin^{3/4} x \cdot \cos^{5/4} x}\, dx$$

$$\int_0^{\pi/4} \frac{1}{\tan^{3/4} x \cdot \cos^2 x}\, dx = \int_0^{\pi/4} \frac{\sec^2 x\, dx}{\tan^{3/4} x}$$

Substitute $\tan x = t$

$$\sec^2 x\, dx = dt$$

$$= \int_0^1 \frac{1}{t^{3/4}}\, dt = \int_0^1 t^{-3/4} dt = 4(t^{1/4})_0^1 = 4$$

8.(1) Differentiate $F(x)$

$$F'(x) = \begin{vmatrix} \sin\alpha & \cos(x+\alpha) & \sin(x+\alpha) \\ \sin\beta & \cos(x+\beta) & \sin(x+\beta) \\ \sin\gamma & \cos(x+\gamma) & \sin(x+\gamma) \end{vmatrix} + 0 + 0$$

$$C_2 \to C_2 + \sin x\, C_1$$

$C_3 \to C_3 - \cos x\, C_1$

$F'(x) = 0$

$\therefore \quad F(x)$ is independent of x

$\therefore \quad F'(x) = 0$

Alternate solution

$$F(x) = \begin{vmatrix} \cos\alpha & \sin\alpha & 1 \\ \cos\beta & \sin\beta & 2 \\ \cos\gamma & \sin\gamma & 3 \end{vmatrix} \times \begin{vmatrix} 0 & \cos x & \sin x \\ x & -\sin x & \cos x \\ 1 & 0 & 0 \end{vmatrix}$$

9.(9) $a = {}^5P_3 = 5 \times 4 \times 3 = 60$

$b = 150$

$$\frac{a \cdot b}{1000} = \frac{60 \times 150}{1000} = \frac{9000}{1000} = 9$$

10.(1) $\dfrac{dx}{dy} = \dfrac{1}{(e^{\sin x} - y)\cot x}$

$$\frac{dy}{dx} = (e^{\sin x} - y)\cot x$$

$$\frac{dy}{dx} + y\cot x = e^{\sin x} \cdot \cot x$$

On multiplying by sin$x$

$$\sin x \frac{dy}{dx} + y\cos x = e^{\sin x} \cdot \cos x$$

$$\frac{d(y\sin x)}{dx} = e^{\sin x} \cdot \cos x$$

$$\int d(y\sin x) = \int e^{\sin x} \cdot \cos x\, dx$$

$$y\sin x = e^{\sin x} + c$$

$$y\left(\frac{\pi}{2}\right) = e + c = e - 1$$

$$c = -1$$

$$y = \frac{e^{\sin x} - 1}{\sin x}$$

$$\lim_{x \to 0} y = \lim_{x \to 0} \frac{e^{\sin x} - 1}{\sin x} = 1$$

11.(3) on Putting $x = 0, y = 0$ in functional equation we will get $f(0) = 0$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{f(x) + f(h) + xh(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(h)}{h} + \lim_{h \to 0} x(x+h) = \lim_{h \to 0} \frac{f(h) - f(0)}{h} + x^2 = f'(0) + x^2$$

$\therefore \quad f'(x) = -1 + x^2$

$\therefore \quad \log_2(f'(3)) = 3$

12.(44) M lies on y-axis

So for M point $x = z = 0$

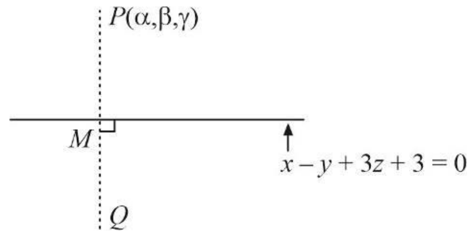
M satisfies the plane $x - y + 3z + 3 = 0$

Hence $y = 3$

M will be (0, 3, 0)

Let the point P is (α, β, γ)

Then Q will be $(-\alpha, -\beta + 6, -\gamma)$



It satisfies $\quad x + y + z = 0$

$$\alpha + \beta + \gamma = 6$$

Also, DR's of PM is $(\alpha - 0,\ \beta - 3,\ \gamma - 0) \equiv (1, -1, 3)$

Hence, $\dfrac{\alpha}{1} = \dfrac{\beta - 3}{-1} = \dfrac{\gamma}{3} = k$

$\alpha = k,\ \beta = 3 - k, \gamma = 3k$

$\alpha + \beta + \gamma = 3k + 3 = 6$

$k = 1$

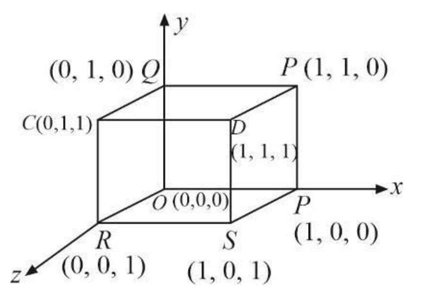
$\alpha = 1,\ \beta = 2, \gamma = 3$

$P(1, 2, 3),\ Q(-1, 4, -3)$

$(PQ)^2 = 44$

13.(1) Let $\quad \hat{l} = l\,\hat{i} + m\,j + nk$ where $l^2 + m^2 + n^2 = 1$

Unit vector along 'd_1' diagonal is $d_1 = \dfrac{\hat{i} + j + k}{\sqrt{3}}$



Similarly $d_2 = \dfrac{\hat{i} - j + k}{\sqrt{3}}$

$$d_3 = \frac{-\hat{i} + j + k}{\sqrt{3}}$$

$$d_4 = \frac{\hat{i} + j - k}{\sqrt{3}}$$

$\therefore \quad [(\hat{l} \cdot d_1)^2 + (\hat{l} \cdot d_2)^2 + (\hat{l} \cdot d_3)^2 + (\hat{l} \cdot d_4)^2]$

$$= \left[\frac{4}{3}(l^2 + m^2 + n^2)\right] = \left[\frac{4}{3}\right] = 1$$

14.(22) $X = \displaystyle\sum_{r=1}^{10} r^2 \cdot ({}^{10}C_r)^2$

$$\sum_{r=1}^{10} r^2 \cdot \left(\frac{10}{r} \cdot {}^{9}C_{r-1}\right)^2 = 100 \cdot \sum_{r=1}^{10} ({}^{9}C_{r-1})^2$$

$X = 100 \cdot {}^{18}C_9$

$$\frac{X}{221000} = \frac{100 \cdot 11 \cdot 13 \cdot 17 \cdot 20}{13 \cdot 17 \cdot 1000} = 22$$

15.(B) $\dfrac{x^2}{x^2-1} \geq 0$

$x^2-1>0$ or $x^2=0$

$\underset{-1 \quad\; 0 \quad\; 1}{+ \;\; - \;\; - \;\; +}$ $E_1 : x \in (-\infty,-1) \cup \{0\} \cup (1,\infty)$

Also, $\dfrac{x^2}{x^2-1} = 1 + \dfrac{1}{x^2-1}$

So, $\dfrac{x^2}{x^2-1} > 1$ or $\dfrac{x^2}{x^2-1} = 0$

$\tan^{-1}\left(\dfrac{x^2}{x^2-1}\right) > \dfrac{\pi}{4}$ or $\tan^{-1}\left(\dfrac{x^2}{x^2-1}\right) = 0$

Domain of $f(x)$ is $E_1 : x \in (-\infty,-1) \cup \{0\} \cup (1,\infty)$

Range of $f(x)$ is $\{0\} \cup (\pi/4, \pi/2)$

For E_2; $\log_{10} \tan^{-1}\left(\dfrac{x^2}{x^2-1}\right)$ is real

$$\tan^{-1}\left(\dfrac{x^2}{x^2-1}\right) > 0$$

$$\dfrac{x^2}{x^2-1} > 0$$

$\underset{-1 \quad\; 0 \quad\; 1}{+ \;\; - \;\; - \;\; +}$ $x \in (-\infty,-1) \cup (1,\infty)$

Domain of $g(x)$ is $x \in (-\infty,-1) \cup (1,\infty)$

Range of $g(x)$ is $(\log_e \pi/4,\ \log_e \pi/2)$

16.(A) There are 8 Boys and 6 Girls

$N_1 = {}^8C_3 \times {}^6C_3 = 1120$

$N_2 = {}^8C_2 \times {}^6C_3 = 560$

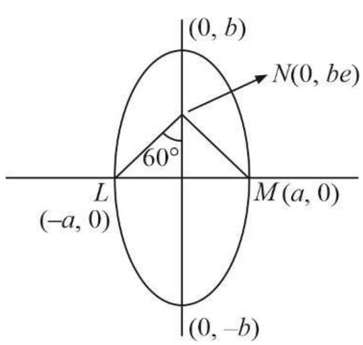
$N_3 = {}^8C_7 \times {}^6C_3 + {}^8C_8 + {}^6C_2 = 175$

(iv) If G_2 and B_3 can't be selected that means we have to select girls from 5 girls and to select boys from 7 boys

Possible ways $= (3 \text{ Boys}, 2 \text{ Girls}) + (4 \text{ Boys}, 1 \text{ Girl}) + (5 \text{ Boys})$

$$= {}^7C_3 \times {}^5C_2 + {}^7C_4 \times {}^5C_1 + {}^7C_5 = 350 + 175 + 21 = 546$$

17.(C) $E : \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ $a < b$



$\dfrac{a}{be} = \tan 60° = \sqrt{3}$

$\dfrac{a}{b} = \sqrt{3}e$ ……(i)

$\dfrac{a^2}{b^2} = 1 - e^2 = 3e^2$

$4e^2 = 1 \Rightarrow e = \dfrac{1}{2}$

Area of $\Delta LMN = \frac{1}{2}\times 2a\times be = \frac{ab}{2} = \sqrt{3} \quad \Rightarrow \quad ab = 2\sqrt{3}$

$\frac{a}{b} = \sqrt{3}e = \frac{\sqrt{3}}{2} \quad \Rightarrow \quad a = \frac{b\sqrt{3}}{2}$

$ab = \frac{b^2\sqrt{3}}{2} = 2\sqrt{3} \quad \Rightarrow \quad b = 2 \quad \Rightarrow \quad a = \sqrt{3}$

(2) Length of minor axis = 2a $= 2\sqrt{3}$

(3) Distance between foci is $2be = 2$

(4) The length of latus rectum $= \frac{2a^2}{b} = \frac{2\times 3}{2} = 3$

18.(B) $f_1(x) = 2\tan^{-1}x$ when $|x| \le 1$

It is continuous and differentiable at $x = 0$

$f_2(x) = 2|x| - 1$

It is continuous but not differentiable at $x = 0$

$$f_3(x) = \begin{cases} \frac{1}{2x+1} & , \quad x > 0 \\ \frac{1}{2x-1} & , \quad x < 0 \\ 1 & , \quad x = 0 \end{cases}$$, It is continuous at $x = 0$

$$f_4(x) = \begin{cases} 0 & , \quad x > 0 \\ 0 & , \quad x < 0, \\ 0 & , \quad x = 0 \end{cases}$$ It is continuous and differentiable everywhere